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Case Study Narratives

Danny Driscoll · Principal Product Designer

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Four case studies: Self-Service Portal · Fraud Prevention Suite · Messaging API Platform · Universal College Application

CASE STUDY 01 · TELESIGN · SELF-SERVICE PORTAL

How Cutting Onboarding From 67 Days to Same-Day Access Freed TeleSign to Serve an Entirely New Market

CONTEXT & STAKES

TeleSign's enterprise onboarding took 67 days on average. Every new customer required manual intervention from the customer success team. Products had to be provisioned by hand. API keys required CS involvement. Government approval documentation across 120-plus countries was handled person by person, email by email.

The business consequences were direct. Deals slipped during the onboarding wait. Cash flow was delayed because revenue only started when customers went live. And the resource cost of handling every onboarding manually made it economically impossible to serve small and medium businesses at all.

Something had to change. I was brought in as Principal Product Designer to redesign the customer onboarding experience from the ground up, during a global pandemic, with a distributed international team and a constrained development budget.

RESEARCH & INSIGHT

I spent a week shadowing the customer success team before writing a single brief. What I saw was a team spending 40 percent of their time answering the same questions repeatedly, provisioning products manually, and managing status updates through back-and-forth email chains. They were not doing high-value work. They were plugging gaps in a broken system.

I then interviewed twelve enterprise customers who had recently completed onboarding. Their frustration was consistent: "I never knew what step came next. I'd wait days for a response." Another said: "I wanted to get started immediately. Instead, I waited weeks."

The insight was not that customers needed better communication from TeleSign. It was that they should never have needed to communicate with TeleSign to complete onboarding at all.

THE DECISION

I designed a self-service portal giving enterprise customers direct control over every step of onboarding: account creation, business verification, product purchasing, phone number acquisition across 120-plus countries, API key generation, and billing — all without CS involvement.

The hardest design problem was not the interface. It was the regulatory variation. Each country had different documentation requirements — some requiring extensive government paperwork, others requiring nothing. The portal had to handle this complexity without exposing it to the customer. I designed dynamic flows that surfaced only the requirements relevant to each customer's location, at the moment they were needed.

Budget was tight. Story points were limited. Every feature required a prioritization decision. I focused ruthlessly on the path from signup to first API call, because that was the moment everything else depended on.

OUTCOME

The results split cleanly across two tiers of complexity, and both tell an important story.

For simpler products and standard configurations, the portal made same-day sign-up possible for the first time. A customer could discover TeleSign, create an account, purchase a product, generate API keys, and make their first API call without ever contacting the CS team.

For advanced configurations requiring government regulatory approval — which vary significantly by country and product — onboarding time dropped from 67 days to 35. Those cases still require process time outside TeleSign's control. What changed was that TeleSign's portion of that process became self-directed, transparent, and fast.

CS workload on onboarding tasks fell 40 percent overall. Eighty-five percent of customers completed onboarding without CS intervention. Daily transaction revenue grew from $500K to over $2M during the period this work was in place. The phone number purchasing UI within the portal won TeleSign's 2019 Innovation of the Year award.

REFLECTION

The biggest lesson was about where design creates leverage. The onboarding flow itself was not particularly complex to design. The complexity was in the back-end regulatory rules for 120-plus countries. Getting those rules surfaced correctly, at the right moment, with the right context, was where the design work had the most impact.

If I were doing this again, I would invest more time earlier in the error state design. When a customer hit a regulatory requirement they were not expecting, the experience dropped sharply. We fixed those cases iteratively, but designing for failure paths with the same rigor as the success path from the start would have shortened that cycle.

CASE STUDY 02 · TELESIGN · FRAUD PREVENTION SUITE

Designing for the Adversary: Making ML-Powered Fraud Prevention Usable for Analysts Making Split-Second Decisions

CONTEXT & STAKES

TeleSign processes over 21 billion transactions annually. The companies relying on that infrastructure include banks, healthcare platforms, gaming companies, and e-commerce platforms serving tens of millions of users. When fraud gets through, the consequences are not abstract. Account takeovers drain real accounts. SIM swap attacks give attackers control of phone numbers tied to financial identities. SMS pumping schemes generate fraudulent traffic that costs clients real money.

TeleSign's Intelligence product analyzed over 1,000 configurable data points to return real-time risk scores on a 0 to 1000 scale, assessing more than five billion unique phone numbers monthly. The product was technically sophisticated. The problem was that the interface for configuring and acting on it was built for engineers, not for the fraud analysts who needed to use it under pressure, in real time, during active attack events.

My job was to make a system of exceptional complexity usable by the people who needed it most.

RESEARCH & INSIGHT

I interviewed fraud analysts across financial services, gaming, and e-commerce clients. Their core frustration was consistent across every sector: the system would flag a user, but the analyst could not understand why. They were being asked to make consequential decisions — block a real customer or allow a potential fraudster — with insufficient information to act confidently.

One analyst said it directly: "I need to understand why the system flagged this user, not just that it did."

That single sentence shaped every design decision that followed. The interface was not failing because the risk model was wrong. It was failing because analysts could not see into the reasoning behind the scores. Confidence in the system collapsed without that visibility.

THE DECISION

I designed the fraud analyst dashboard to surface the specific signals driving each risk score — not just the score itself. SIM swap detection, breached data matches, porting history, location anomalies, carrier inconsistencies — all made visible and scannable at a glance.

I also designed the configuration layer for non-technical fraud analysts to adjust risk thresholds by attack type, by country, and by client context, without writing code or filing an engineering ticket. The system supported over 1,000 configurable parameters. The design challenge was exposing the right controls to the right people without overwhelming them.

For the Ticketmaster concert promotion work specifically, I built an admin interface handling country-by-country eligibility for millions of simultaneous applicants, with real-time risk scoring, location verification through carrier network data, and budget controls preventing runaway promotional costs. The user-facing experience communicated location verification to legitimate fans without creating anxiety, using clear language about why verification protected their access.

OUTCOME

The combined Intelligence and PhoneID suite protected over 21 billion annual transactions. The SMS Country Blocking feature — part of the same design system — delivered an 85 percent reduction in spam complaints for one financial services client within the first month of deployment, and mitigated $900K in negative margin exposure from SMS pumping attacks.

The Ticketmaster promotion delivered millions of secure redemptions while blocking bot-driven abuse, VPN masking, and location spoofing at scale.

REFLECTION

Designing for security experts forced a level of precision I had not previously encountered. Fraud analysts are skeptical of systems they cannot interrogate. Earning their trust required making the machine's reasoning transparent, not just its output.

The deeper lesson was about the cost of false positives. Blocking a genuine Ticketmaster fan to protect against fraud is not a neutral outcome. Every threshold decision had two failure modes, not one. Good fraud prevention design holds both in view simultaneously.

CASE STUDY 03 · TELESIGN · MESSAGING API PLATFORM

Six Channels, One Interface: Unifying Enterprise Messaging Without Losing What Made Each Channel Valuable

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CONTEXT & STAKES

Enterprise clients needed to reach customers across six different messaging channels. SMS, MMS, RCS, WhatsApp, Viber, and email each had separate APIs, different authentication requirements, different media rules, and different regulatory constraints across 120-plus countries. One Fortune 500 e-commerce client had spent six months integrating five separate messaging vendors. When a message failed on one channel, there was no automatic fallback. Customer notifications arrived late or not at all.

The cost was not just operational. Compliance exposure was real: TCPA violations ran $500 to $1,500 per message. GDPR violations could reach four percent of annual revenue. With messages going to customers in dozens of countries, each with different rules, the legal surface area was enormous.

TeleSign needed one interface for all six channels. I designed it.

RESEARCH & INSIGHT

I interviewed three distinct user groups: the developers integrating the API, the marketing teams building message templates, and the compliance officers managing regulatory risk across country markets.

Each group had a different relationship with the same underlying complexity. Developers wanted direct API control and clear documentation. Marketing teams wanted visual tools that did not require them to learn JSON. Compliance officers wanted guardrails that prevented violations before they happened, not reports after.

The critical research finding came from a financial services client during testing: "We need to know exactly why a message failed. Was it a carrier issue, a compliance block, or a user opt-out?" They were not asking for more data. They were asking for clarity in failure states — which were the moments that mattered most.

I also researched RCS closely. It was about to launch on iOS 18, giving early adopters a significant advantage. RCS delivered 22.2 percent click-through rates versus three percent for standard SMS. The platform needed to be built for where messaging was heading, not where it had been.

THE DECISION

The core design challenge was a single create-once, deploy-everywhere template system. Marketing teams needed to build one template and have it render correctly across all six channels — with the richer channels showing full media and interactivity, and the simpler channels gracefully degrading to clean text.

I built a visual template builder with real-time preview rendering across all six channels simultaneously. Non-technical users never touched JSON. The system generated it automatically from the visual input.

For fallback configuration, I designed a drag-and-drop priority ordering system. Clients set their channel cascade: RCS first, WhatsApp second, SMS as the floor. Timing between fallback attempts was configurable. Geographic rules were surfaced inline — because WhatsApp does not operate in China, RCS availability varies by carrier, and those constraints had to be visible at the moment decisions were made.

Compliance was embedded into template creation, not bolted on afterward. Regulatory requirements for the destination country appeared in real time as templates were built, stopping violations before they were committed rather than flagging them after.

O U T C O M E

Enterprise implementation time dropped 50 percent. Clients previously spent four to six months integrating multiple messaging vendors. The unified API reduced that to two to three months.

Message delivery rates improved 15 to 20 percent through intelligent channel cascading. RCS campaigns for early adopters achieved 22.2 percent click-through versus three percent for standard SMS.

The platform contributed to 42 percent annual revenue growth, supporting TeleSign's growth to a $1.3 billion valuation at acquisition.

R E F L E C T I O N

The most important design decision in this project was embedding compliance into the creation flow rather than the review flow. Every alternative I considered would have surfaced regulatory issues after teams had already built their templates. That would have created rework, frustration, and eventually workarounds that bypassed the guardrails entirely.

Making constraints visible early is not just a usability principle. In high-compliance environments, it is a risk management decision. The design of the warning is part of the product.

CASE STUDY 04 · CAPPEX (NOW APPILY.COM) · UNIVERSAL COLLEGE APPLICATION

From One Student's Nightmare to 4 Million Applications: Redesigning College Admissions at Scale

CONTEXT & STAKES

In 2015, the college application process punished the students who could least afford it. Applying to ten schools meant writing ten applications, tracking ten deadlines, and entering the same personal information ten times over. At schools charging $50 to $75 per application, the financial barrier alone stopped many students from applying broadly enough to find the right fit or the best financial aid offer. The average completion rate across the industry sat between 20 and 35 percent. Students were not lazy or unprepared. They were priced out and worn down.

First-generation applicants faced the sharpest version of this problem. No family member had navigated this before. The costs were real. The system assumed a familiarity and a financial cushion they had not been given. Many stopped applying entirely. The first-generation college graduation rate at the time was 24 percent, versus 59 percent for continuing-generation students. That gap had a design problem at its root.

Cappex (now Appily.com) asked whether one platform could remove those barriers for good. I led the research and design work that answered that question. The research, the university partnerships, and the product decisions made during my tenure became the foundation Appily was built on after EAB's acquisition in 2020.

RESEARCH & INSIGHT

I spent months inside the problem before touching a single wireframe.

I ran surveys, interviews, and focus groups with high school students across the country. I watched them attempt to complete multiple applications simultaneously. They had dozens of browser tabs open. They copied and pasted the same essay fragments between windows. They missed deadlines. They gave up mid-submission.

One student told me: "I'm applying to eight colleges. I've written twelve different essays. I'm exhausted and I'm scared I'll send the wrong thing to the wrong school." That was not a workflow problem. That was a dignity problem.

I also met directly with admissions offices at partner universities. What they told me reframed everything. They wanted more applicants. They were losing qualified students not to competing schools but to the friction of applying. Universities and students were failing each other because the system connecting them was broken.

The insight was this: students did not need a better application. They needed to apply once.

THE DECISION

I designed the Universal College Application. One essay. One submission. Accepted by 200-plus partner institutions simultaneously. The tagline came from the research: "One Essay, Every College."

This was not a simplification of the existing process. It was a replacement of the underlying logic. Instead of asking students to adapt to each institution's requirements, I worked with partner admissions offices to build a universal standard they could accept. That required as much relationship design as interface design.

I chose to remove features rather than add them. Early testing showed students wanted less to manage, not more control. Every decision was made against a single question: does this reduce the barrier, or does it add one?

For a significant portion of partner universities at launch, the application was free. Removing the per-application fee directly addressed the financial barrier that stopped first-generation students from applying broadly in the first place.

OUTCOME

The Universal College Application achieved a 47 percent completion rate. The industry standard was 20 to 35 percent.

A student applying to ten schools spent six hours on the Cappex platform. Traditional applications required more than thirty hours for the same outcome. And with free access at launch for many partner schools, the financial barrier that had stopped first-generation students from applying broadly was removed entirely.

The platform grew from 250,000 users in 2015 to 1.5 million by the end of my tenure in 2018. When EAB acquired Cappex and relaunched it as Appily.com in 2020, the user base reached four million. The research, the design decisions, the university partnerships, and the free-access model built during those years became the foundation Appily was built on.

REFLECTION

The metric I am most proud of is not the completion rate. It is the students who applied to schools they would not have applied to otherwise because the process was no longer exhausting enough to stop them.

If I were doing this today, I would push harder on accessibility earlier. We optimized for completion. We should have measured, from day one, whether first-generation students were completing at the same rate as their peers. That data would have changed some of our prioritization decisions.

The other thing this project taught me: institutional buy-in is a design problem. Getting 200 universities to accept a universal essay prompt required understanding their constraints as deeply as I understood students'. The best product decisions in that project came from those admissions office conversations, not the wireframes.